Understanding 5G Performance for Real-world Services: a Content Provider's Perspective

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ABSTRACT

5G has seen rapid growth recently, attracting several measurement studies on its coverage, connectivity and quality of service. However, there is still a lack of understanding of 5G's capabilities and potential impacts from a *content provider (CP)*'s perspective. This paper fills in this gap by studying 5G networks used by over 23 million users in one year in Kuaishou, a popular *crowdsourced live streaming* platform. Our measurements provide the following discoveries. i) Standalone (SA) 5G generally provides end-to-end performance improvement as compared with 4G or non-SA (NSA) 5G, but its advantage depends on both the number of cellular users and CP-level configurations. ii) In the radio access network, SA 5G is more sensitive to access density but has better handover tolerance. iii) Controlled experiments with 29 mobile device models on energy consumption refute some "conventional wisdom," including that 5G always consumes more power. iv) Traceroute-based active experiments in over 300 cities show that although users are "closer" to the internet in SA 5G, their end-to-end latency may not benefit from that. Furthermore, we show new design space for 5G participants and provide a 5G-aware rebuffer strategy tested by 9 million viewers in Kuaishou, with a 7% reduction in rebuffer proportion.

CCS CONCEPTS

• **Networks** → **Mobile networks**; **Network measurement**; *Network performance analysis*; Network reliability; • **Information systems** → **Multimedia streaming**;

KEYWORDS

5G, SA 5G, End-to-End Performance, Radio Access Network, Core Network, Content Provider, Crowdsourced Live Streaming

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1 INTRODUCTION

Recent years have witnessed a rapid growth of both 5G base station deployment and end users. Since the first full set of 5G New Radio (NR) standards (3GPP R15) was completed [5], 5G base stations have been deployed all over the world, to which only China has contributed over 1.15 million 5G base stations [29]. According to Cisco Annual Internet Report [13] and Ericsson Mobility Report [16], by 2023, 5G connection will have a bandwidth improvement by 13 times, and by 2027, 5G will account for 49% of all mobile subscriptions.

As 5G is promised with higher bandwidth, lower latency, better reliability and massive access support compared with 4G [5, 6], it has attracted several measurement studies on its coverage, connectivity and quality of service (QoS), as well as new applications to be enabled by it, such as bandwidth-hungry UHD and 360/VR video [42], latency-sensitive auto-driving [12] and massive access of IoT (Internet of Things) devices [38].

For the measurement studies, Xu *et al.* [44] and Narayanan *et al.* [31] conducted measurements on 5G coverage and radio access characteristics, mainly from the perspective of internet service providers (ISPs). Narayanan *et al.* [33] also invested throughput, power consumption and quality of experience (QoE) implications for users, and proposed a 4G/5G switching scheme to save energy. Li *et al.* [27] investigated 5G reliability from the perspective of a smartphone vendor, and proposed a recovery strategy to improve network reliability.

On the one hand, previous measurement studies mainly focus on NSA 5G, which still uses 4G's core network. For the SA 5G, Narayanan *et al.* [33] took a look at the end-to-end delay and throughput of low band SA 5G (*i.e.*, 600MHz). They showed that SA 5G performed worse than NSA 5G, which might be because of the immaturity of SA 5G at that time. Therefore, the mysteries about SA 5G's are still remaining to be fully revealed. With the increasing deployment of middle band SA 5G (*i.e.*, 2.6GHz and 3.5GHz) in China, our work distinguishes itself with previous ones partly on

that we mainly focus on SA 5G and are the first to take a deep dive into the 5G core network.

On the other hand, previous measurement studies are mainly carried out from the perspectives of ISPs and smartphone vendors; there is still a lack of understanding of 5G's capabilities and potential impacts from a *content provider (CP)*'s perspective. There are open questions for a content provider to address before unleashing the power of 5G: 1) From a content provider's perspective, what QoS/QoE benefits can a CP gain from the increasing usage of 5G networks? 2) What practical strategies can be taken by a CP to fully utilize 5G's new features on both performance and architecture?

This paper tries to answer these questions by studying 5G networks used by over 23 million users in a time span of one year (December 2020–November 2021) in Kuaishou, one of the largest crowdsourced live streaming (CLS) content providers. Crowdsourced live streaming, as one of the killer applications for mobile networks, is expected to see even faster growth with 5G networks. In fact, Kuaishou has a fraction of 5G users (users with 5G devices) increasing from 13.4% in December 2020 to 32.1% in November 2021.

In a crowdsourced live streaming system, broadcasters upload video streams to source servers, which will then distribute the streams to millions of potential viewers. Understanding 5G's potential from a content provider's perspective actually raises technical challenges.

First, it is non-trivial to conduct measurement in such a large and complex online service system. i) Although there are many performance indicators for crowdsourced live streaming, most of them are less useful for studying 5G networks directly, *e.g.*, stream bitrates. It requires a set of metrics that not only are representative enough in crowdsourced live streaming, but also reflect the characteristics of 5G networks. ii) There are a huge number of conditional factors, *e.g.*, device models and encoding modes, of which different combinations may have diverse impacts on the above metrics. However, it is impossible to enumerate and record all of them, making it hard to obtain "unbiased" results. iii) As data collection consumes additional resources on user devices, it is required to minimize the impact on users' QoE while collecting the data. As an online service serving *hundreds of millions of users* per day, any possible negative impact on the application stability should be avoided.

Second, third parties involved in this system challenge both the data collection and analysis. i) To distribute streams, Kuaishou utilizes content delivery networks (CDN) from multiple providers, which often act as black boxes in the stream delivery, affecting our analysis on many metrics, *e.g.*, broadcaster-to-viewer delay. ii) Internet service providers (ISPs), which also act as mobile network operators (MNOs) in China, seldom open the configurations and deployment information of the core network to the public, and it requires additional efforts to make sense of the data collected by a content provider, to reveal 5G network characteristics.

To overcome these challenges, we have designed and implemented the following data collection mechanisms: *joint active and passive data collection* to validate data from different app and OS versions, and *private field insertion* to overcome the third parties' impact. With carefully-designed experiments, we answer the questions above and provide insights for content providers. Our key contributions and major discoveries are summarized as follows.

▷ To the best of our knowledge, we are the first to provide a large-scale measurement from a CP's perspective, on both QoS provided by 5G networks and users' QoE under 5G in crowdsourced live streaming. We measure end-to-end performance at both broadcasters and viewers, as well as the overall *liveness* from broadcasters to viewers. We find SA 5G provides reduced application-level latencies, *i.e.*, ∼60% of 4G's connection establishment delay at broadcasters and 67∼89% of 4G's overall transmission delay, and higher download speed. Besides, we present SA 5G's better performance over NSA 5G, a different result from that in the US [33]. However, SA 5G's download speed shows a decreasing trend as the number of SA 5G users increases.

▷ We investigate the radio access network (RAN) performance. Measurements on users in Beijing and Shenzhen show that SA 5G is more sensitive to access density than 4G, especially when the access density is at a low level. Results on handovers show that horizontal handovers are much more frequent than vertical handovers and both kinds of handover can highly affect users' QoE. For vertical handovers, those between NSA 5G and 4G have the least impact on rebuffer frequency; while for horizontal handovers, those within SA 5G have the least impact on rebuffer frequency.

▷ Energy consumption is also investigated. Based on measurements of 22 million users with 27 Android and 2 iOS 5G-capable device models, we examine several pieces of conventional wisdom on energy consumption, and show that some conventional wisdom does not always hold true. Using SA 5G is not necessarily more power-consuming than using 4G: 7 out of 29 device models are less power-consuming in SA 5G than in 4G. And the energy performance is highly related to 5G SoCs: the gaps between SA 5G and 4G for Kylin and Dimensity range from −5% to 6.5%, but for Snapdragon and A14, the gaps range up to 18% and 17%, respectively.

▷ We explore the impact of 5G core network. Based on 1.7 million users' traceroute results, we find that users in SA 5G are "closer" to the internet. However, being closer to the internet does not guarantee a reduced user-to-server RTT; instead, a shorter gateway-server distance is usually more helpful to reduce the RTT.

▷ We also provide insights and system optimization guidelines for 5G participants, including content providers, ISPs, *etc.*, with respect to server placement strategy, routing policy, *etc.* Based on the measurement insights, we propose a 5G-aware rebuffer strategy tested by 9 million viewers in Kuaishou, with a 7% reduction in rebuffer proportion.

2 BACKGROUND

In this section, we present how 5G networks are used by a crowdsourced live streaming content provider.

2.1 Crowdsourced Live Streaming

In a crowdsourced live streaming system, any individual can act as a *broadcaster* (or *uploader*), who creates a live *channel* and constantly uploads the video stream to this channel, usually from their smartphones. Other users are then able to join the channel as *viewers*, and receive the stream from content delivery servers. Different from traditional video on demand (VoD) service, in crowdsourced live streaming service, viewers can interact with broadcasters in real time. Thus, crowdsourced live streaming is expected to benefit

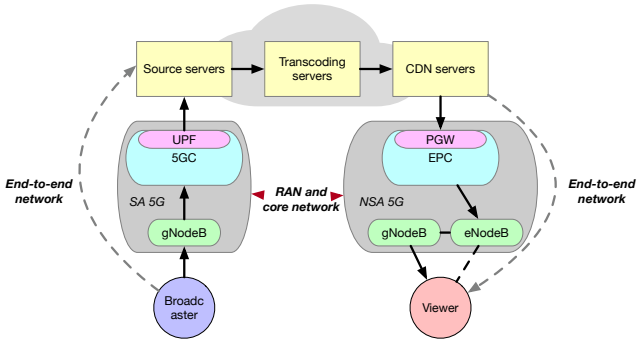


Figure 1: Illustration of a crowdsourced live streaming system. Note that in this figure we only plot users in 5G networks, and many other users are still using 4G or Wi-Fi networks.

more from 5G's low latency. Popular crowdsourced live streaming systems such as Twitch [14] and Kuaishou [24] today are able to attract hundreds of millions of active users every day. The quality of experience for both broadcasters and viewers is usually determined by the quality of cellular or Wi-Fi networks they are connected to.

2.2 5G Networks

5G networks include *non-standalone* (NSA) and *standalone* (SA) deployments [18], whose structures are illustrated in Fig. 1. To utilize the legacy 4G core network (Evolved Packet Core, EPC), the NSA 5G only deploys 5G base stations (gNodeB) and shares the core network with 4G. It requires a UE to connect to both a 4G base station (eNodeB) for the control plane, and a 5G base station (gNodeB) for the user plane [18]. On the other hand, SA 5G deploys both gNodeB and 5G core network (5GC), indicating that a UE only establishes a single connection with gNodeB. As for the core network, the gateway between 4G network and the internet is a PDN gateway (PGW), usually placed only in the capital cities in China. With the control and user plane separation (CUPS) architecture in 5GC, the gateway, which is called user plane function (UPF), can be placed much closer to users [30, 50].

As SA 5G requires more infrastructure deployment, ISPs usually initiate with NSA 5G and gradually migrate to SA 5G. Currently, the three major ISPs in China, China Mobile (or Mobile hereinafter), China Unicom (or Unicom) and China Telecom (or Telecom), have both NSA and SA 5G deployments [17, 19, 28]. In particular, for the access networks, different radio bands are utilized: China Mobile uses the 2.6GHz band (n41), while China Unicom and Telecom co-construct the base stations and both use the 3.5GHz band (n78) [8].

2.3 How 5G is Used by a CLS CP

Next, we present the general architecture of a crowdsourced live streaming system with users in 5G networks, using Kuaishou system as an example. In today's 5G network deployment, a 5G-deviced broadcaster/viewer can be either connected to SA 5G or NSA 5G, as illustrated in Fig. 1. After being connected to a *source server*, the broadcaster will generate the stream and upload the stream over streaming protocols such as real-time messaging protocol (RTMP) to the source server. To support multiple bitrates, the stream can also be *transcoded* into a set of bitrate versions on the *transcoding servers*. After that, the streams are replicated to many CDN servers in different regions, to support the scaling volume of viewers, who will receive the stream from these CDN servers[1].

Table 1: Statistics of the dataset.

# of	Wi-Fi+Cellular	Cellular	SA 5G
viewer logs	1.78×10^{12}	340×10^{9}	13.1×10^{9}
broadcaster logs	26.7×10^{9}	6.26×10^{9}	406×10^{6}
users	$\geq 260 \times 10^{6}$	$\geq 170 \times 10^{6}$	$\geq 23.0 \times 10^{6}$
sessions	316×10^{9}	48.0×10^{9}	2.75×10^{9}
streams	124×10^{6}	48.5×10^{6}	3.44×10^{6}
device models	$\geq 17 \times 10^{3}$	$\geq 10 \times 10^{3}$	≥ 500

Though 5G networks have already been deployed all over the world, there are still several mysteries for content providers, which will be studied in this paper. 1) Do users connected to 5G networks always have better quality of experience than users connected to other mobile networks, and how much benefit can users gain from using 5G? 2) What are the factors that may affect the application-level quality of experience for users under 5G? 3) To fully utilize 5G's potential, what concerns should be considered and what actions should be taken for a content provider, when more users in the system are using 5G networks. This paper answers these questions from the end-to-end network, radio access network and core network perspectives.

3 MEASUREMENT METHODOLOGY

We build passively-collected large-scale datasets as well as carry out active experiments in our measurement studies. Our measurements have been inspired by several previous studies, including the extensive measurement of 4G network performance on high-speed rails [41], traceroute-based network topologies analysis in 4G [39, 48], and several measurement studies on 5G networks [31, 44].

3.1 Passive Data Collection

In Kuaishou, both broadcasters and viewers use the same app to upload and receive the video streams. We have implemented data collection mechanisms in the app for both broadcasters and viewers to continuously collect and periodically upload information to the *log servers*. In particular, status information and statistics at a user in the past time window (*i.e.*, 10 seconds), which is also called a *slice*, are reported at the end of each slice. Broadcasters and viewers have quite similar reporting mechanisms: trace items are both generated and uploaded every slice.

In each slice, for both broadcasters and viewers, the *status information* was collected, including 1) the device information, *i.e.*, the user's device model, ID and battery level (*e.g.*, 80 (percent)), 2) the streaming and session information, *i.e.*, the channel/session ID and video codec information, and 3) the network information, *i.e.*, the

[1]There are two kinds of streams for Kuaishou: single-bitrate streams and multi-bitrate streams. They do not only differ in the way that viewers' phones treat them (*i.e.*, adaptive bitrate (ABR) strategy is adopted for the latter), but also in the transmission paths (*i.e.*, the former streams will bypass the transcoding servers). To ease the burden of understanding the complex system and help readers focus on 5G and live streaming, we will only mention these two streams when making comparisons between them.

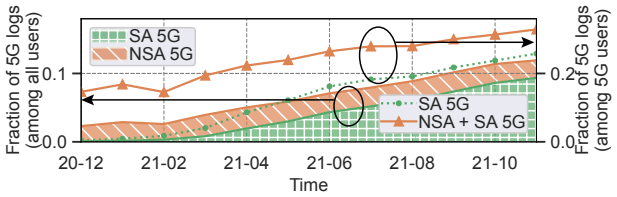


Figure 2: Evolution of fraction of 5G logs.

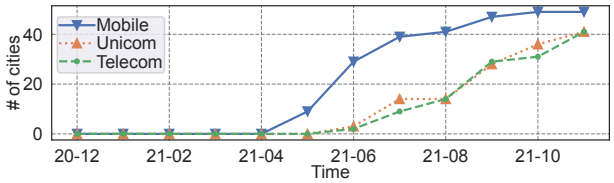


Figure 3: Number of cities whose fraction of SA 5G logs among 5G users exceeds 25% (among 49 representative cities).

network type (*e.g.*, "SA 5G"), ISP (*e.g.*, "China Mobile"), the server's IP, and the base station information[2].

The QoS/QoE information was also collected. To make the measurement results useful and more reflective of the network performance rather than CPU or device performance, we try to strip out the most network-relevant parts from QoE metrics, so as to make them less app-/device-specific. For example, for the broadcasters' QoE, we do not explicitly present the startup delay, as it is highly app-specific and device-specific: applications of different versions may load different local resources for displaying, and more powerful CPUs or devices generally spend less time on startup. Instead, we collect the connection establishment delay, *i.e.*, the delay for establishing an RTMP connection, which is a major component of the startup delay. Note that the live streaming is much different from video on demand, where the former also stresses the real-time interaction between broadcasters and viewers. Therefore, besides rebuffer duration and download speed, we also collect the broadcaster-to-viewer delay as a QoE metric. Here the download speed is that of I-frames, since the download speed in the whole slice is more constrained by video bitrates than network capacities. More details will be discussed in the corresponding sections.

In our passive data collection from December 2020 to November 2021, we have collected over 100TB raw logs from both broadcasters and viewers. We have totally 1.78×10^{12} logs (in which 1.31×10^{10} are labeled with SA 5G) reported by viewers and 2.67×10^{10} logs (in which 4.06×10^{8} are labeled with SA 5G) reported by broadcasters. More details are illustrated in Table 1.

Limitations of Data Collection. 1) The CellLocation API *does not always return valid base station information*. However, by a small-scale field test, we find that once it returns the valid values, they are accurate and true. Thus, it is still safe to use it to detect handovers. 2) The granularity of data collection is 10 seconds and there is only one network type (and base station) labeled for each slice. Thus, *we are not able to detect network type (or base station) change happening within a slice*. However, since such in-slice change is very rare, it makes little if any impact on our measurement results and conclusions to be drawn.

3.2 Active Measurement

Inspired by [39, 48], we further study the core network's structures by conducting an active measurement with traceroute. We first select a set of 2 domain names of source servers and 2 domain names of CDN servers, and use them as the destinations for running traceroute tasks. We then select the most active ~5 million Kuaishou 5G users (Android users only) and assign the traceroute tasks to those who are online at around 7pm on two days in September 2021. Each user executes the traceroute 5 times to reduce randomness, and we have successfully retrieved traceroute results from over 86.6k users and 87.7k users on the two days separately. Among the results, 10.6% are labeled with SA 5G and 32.7% are labeled with 4G.

Although the 5G and 4G core networks are IP-based, ISPs have specific configurations on the core network routers and build special tunnels, and a certain fraction of the routers in the core network may not respond to the ICMP request in the traceroute tasks. Therefore, we do not dive into all routers in the core network, but only focus on the first internet hop, which will be approximated as the "gateway" in §6.

According to [48], the traceroute-based methods will not always work in countries where the access and retail functions of ISPs are separated. In these scenarios, the subscriber traffic may be tunneled by an access provider to some entity where the retail provider is co-located, and the detected structure can be meaningless. In our study, we do not encounter such problems, but since we are not totally clear whether China Unicom and Telecom use additional strategies to deal with the co-constructed infrastructure, we mainly focus on China Mobile in the active measurement.

3.3 Overview of 5G Network Usage

Before we present 5G's performance, we first show the fraction of 5G logs in a 12-month time span in our study. According to the *network type* label in the logs reported by both broadcasters and viewers, we are able to calculate the fractions of SA 5G and NSA 5G logs among cellular logs respectively for each month, and we plot these fractions over time in Fig. 2. We observe that the NSA&SA 5G logs have increased from 2% in December 2020 to 12.6% in November 2021 in our system. Among all the 5G users (*i.e.*, users with 5G devices), the SA 5G logs have improved from 0.4% to 25.8%, and the NSA 5G has decreased from 14.2% to 7.1%, indicating that more effort has been put into building SA 5G base stations. Note that there is a slight drop of the fraction in February 2021, during the Chinese New Year holiday, and we suggest the reason is that many users are migrating for family gatherings from metropolis to suburb areas where 5G networks are under deployed at that time.

We also investigate how 5G networks are used in different cities in the same time span. In our study, we select 49 most developed cities of the first three "tiers" in China [2], and use them as samples to study the 5G usage and deployment situation. In Fig. 3, we plot

[2]Base station information includes mobile country code (MCC), mobile network code (MNC), location/tracking area code (LAC/TAC) and cell ID (CID), and is only available for Android devices in our study. We find that the new CellIndentityNR class [9] deteriorates users' quality of experience due to its incompatibility with some devices, and thus use the old class CellLocation [10] to retrieve the base station information.

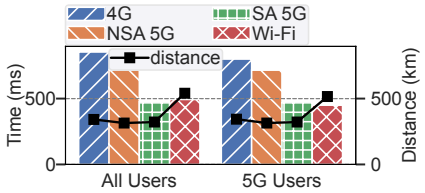


Figure 4: Connection establishment delays and user-server distances for Android users.

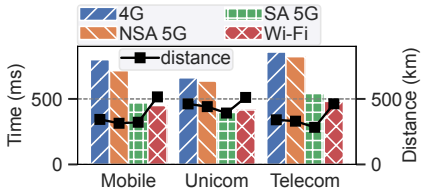


Figure 5: Connection establishment delays and user-server distances in different ISPs.

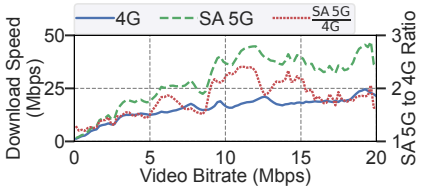


Figure 6: Download speed and SA 5G to 4G gap versus video bitrate.

the number of these cities whose fraction of SA 5G logs among 5G users is above a threshold of 25%, (*i.e.*, 5G users have a chance of over 25% to use SA 5G) of the three major mobile ISPs in China, *i.e.*, China Mobile, China Unicom and China Telecom, over the time span of 12 months. We find that the number of cities has reached 49 for China Mobile by October 2021, but only 41 for China Unicom and Telecom by November 2021, indicating China Mobile deployed SA 5G infrastructure faster than China Unicom and Telecom. As China Unicom and Telecom co-construct 5G base stations and share similar radio bands, their curves are very similar.

This rapid increase in 5G's usage and deployment provides us the opportunity to investigate the performance of SA 5G, especially that of its brand new core network, and make comparisons with NSA 5G and 4G.

Ethical Concerns. The data collection and analysis in this study comply with the agreement established between Kuaishou and its users. No personally identifiable information (*e.g.*, phone number, IMSI) was collected and the collection of base station and location information requires users' additional permission action. We never and have no way to associate the collected data with the users' real identity. The analysis was conducted within Kuaishou's platform which avoids data leakage.

4 END-TO-END PERFORMANCE

In this section, we investigate the "end-to-end" performance in crowdsourced live streaming in both cellular networks and Wi-Fi, covering the broadcaster-to-server, server-to-viewer, and broadcaster-to-viewer performances. We study them from three perspectives: delay, bandwidth and reliability. We have found that some performances in three ISPs (*resp.*, months) are similar or lead to the same conclusions. Therefore, to avoid redundant discussion, we will only present the results of users in China Mobile (*resp.*, September) when no comparison is involved between ISPs (*resp.*, months).

4.1 Broadcasters to Servers: Reduced Delay

We first study the performance from broadcasters to source servers, especially focusing on the delay. To start a live streaming channel over RTMP, an RTMP connection [1] has to be first established between the broadcaster and a source server. During the establishment, the user and the server first establish a TCP connection, and then they have several handshakes over TCP and negotiate by exchanging necessary variables to establish an RTMP connection. We measure the whole delay for RTMP connection establishment, which is a major component of startup delay and would be much higher than RTT.

We first show that the highly network-dependent delay is also impacted by the devices' computational ability. In Fig. 4 we present the average connection establishment delays in different networks, measured among all users (users having used Kuaishou app) and 5G users (Kuaishou users with 5G devices). We find that in 4G and Wi-Fi, the delays among all users are higher than those among 5G users (by ~50ms). We calculate the average distance between users and servers (city-level distance), which is also shown in Fig. 4. We find that the user-server distance averaged among all users and the distance averaged among 5G users are nearly the same for each network type. Thus, we blame the delay differences in 4G and Wi-Fi between all users and 5G users mainly on the devices' different computational ability: 5G devices are generally newer and equipped with more powerful CPUs, which can deal with the negotiation process faster and time the establishment more accurately than old 4G devices. Therefore, we conclude that although the delay is a highly network-dependent metric, the low computational ability of 4G devices may extend the measured delays. To make fair comparisons among different networks, we will only focus on 5G Android users in this section.

With fair comparisons, we find that SA 5G has better delay performance than 4G. In Fig. 5, we present the average connection establishment delays in different ISPs, and the average user-server distances as well. We find that in all the three ISPs, SA 5G's delays are much lower than those in 4G and NSA 5G (by 33%~40%), although the user-server distances are similar in all cellular networks. SA 5G's superiority is consistent with our expectation. However, we find that delays in NSA 5G are only slightly lower than those in 4G (by 4%~10%). This is because that NSA 5G uses 4G's core network and only reduces the delay in the air interface, which is much lower than that in the core network and internet backbone.

Besides, although China Unicom and Telecom share some infrastructure, the delays in SA 5G differ greatly from each other among ISPs. Note that the rank of user-server distances among different ISPs does not match that of delays. For example, in SA 5G, China Telecom has the shortest user-server distance, but the highest delay, while China Unicom shows the opposite. This means that the different delay performances in SA 5G among different ISPs are not because of different user-server distances, but other factors such as the core network deployment.

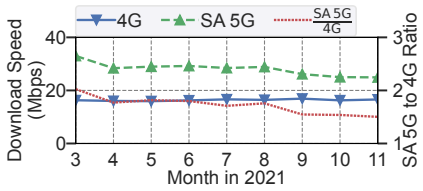


Figure 7: Download speeds in 9 months in 2021.

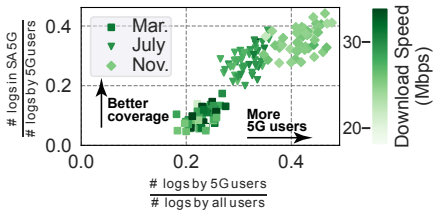


Figure 8: Evolution of SA 5G download speeds, coverage and users in 49 representative cities in 2021.

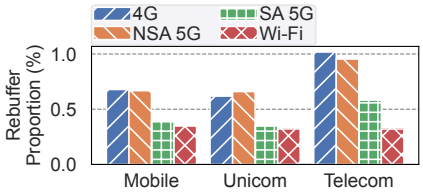


Figure 9: Rebuffer proportions for sessions without changes on network type.

4.2 Servers to Viewers: Improved Bandwidth and Reliability

From CDN servers to viewers, we focus on download speeds and rebuffers at viewers, which reflect SA 5G's bandwidth utilization and reliability.

Download Speed. As we have explained in §3.1, fair comparisons should be made among streams with similar bitrates. In Fig. 6, we plot the average download speeds under different video bitrates. In general, we observe that the download speeds in SA 5G are much higher than those in 4G under all video bitrates, but still much lower than the theoretical capacity [20]. On the other hand, the download speeds in both 4G and SA 5G improve as the bitrate increases. However, the improvement is different for 4G and SA 5G; in particular, we plot a reference curve as the fraction of the download speed of SA 5G over the download speed of 4G (*i.e*, $\frac{\text{SA 5G}}{\text{4G}}$), and observe that the curve increases from ~1× to over 2×, suggesting that SA 5G's download speed improves more than 4G's as users watch streams with higher bitrates. These results are also consistent with observations in previous studies [22, 25]: the achievable download speed depends on the characteristics of the transmission flows and transmission protocols.

Further, we are curious about whether SA 5G's higher download speeds are because it had fewer subscribers at that time or because of its advanced technology. We plot the average download speeds from March 2021 to November 2021 in Fig. 7. In this figure, we choose the stream bitrates to be 6~7Mbps, which are adequate for high frame rate 720p videos. We observe that download speeds in 4G are stable at around 16Mbps, while both download speeds in SA 5G and the reference curve show the trend of decreasing. As the 5G users and usage increase greatly in 2021 (§3.3), this indicates that SA 5G's speed and advantages over 4G deteriorate as SA 5G gets more users.

However, this result can be impacted by biased distribution of 5G users and deployment: a rapid increase in the number of 5G users in where the internet's intrinsic performance is worse than the national average, *e.g.*, in some remote districts, may deteriorate the nationwide performance. To avoid its potential impact, we select 49 cities (same as in §3.3) as samples and plot their download speeds in SA 5G in March, July, and November in 2021 in Fig. 8. The x-axis is the ratio of number of cellular logs uploaded by 5G users to that by all users, which is positively related to the number of 5G users and their activity. The y-axis is the ratio of number of cellular logs in SA 5G to cellular logs uploaded by 5G users, which is positively related to the coverage of SA 5G. In this figure, we observe that the download speeds of SA 5G show the trend of decreasing as x increases and the coverage expands, in nearly all of these cities. Therefore, we conclude that *fewer users using SA 5G partly contribute to its advantages over 4G*. The impact of users' density is further investigated in §5.1.

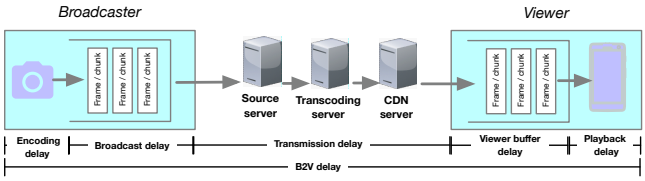


Figure 10: Delays involved in a full channel.

Rebuffers. As a critical factor for viewers' QoE, a rebuffer happens if the playback has consumed all the frames buffered at the viewer's cache, usually caused by the download speed drop or communications outage. The rebuffer proportion, which is defined as the fraction of rebuffer duration over the session's duration, reflects the reliability of connections. In Fig. 9, we plot the average rebuffer proportions for *sessions without changes on the network type*. We observe that *SA 5G has much lower rebuffer proportions*, nearly half of those in NSA 5G and 4G. Besides, NSA 5G users can even have a larger rebuffer proportion than 4G users in China Unicom, meaning that *NSA 5G does not obviously enhance the reliability*. Hence, we conclude that the reliability is more related to the control plane than to the user plane, as NSA 5G still uses 4G base stations for the control plane.

Besides, as three ISPs use different technologies (*i.e.*, LTE-TDD and LTE-FDD [3, 4]) and radio bands in 4G, the different rebuffer performances for three ISPs in 4G are expected. However, it's interesting to find that the rebuffer proportions in SA 5G also differ greatly in China Unicom and China Telecom, although they share the same radio access network infrastructure and similar radio bands. As the two ISPs have similar rebuffer proportions in Wi-Fi, indicating that the internet backbone is not the bottleneck to account for China Telecom's worse performance in cellular networks, we infer that China Telecom's core network might be to blame (*e.g.*, Access and Mobility Management Function involved in the handover procedure [7, 35]). We will further investigate the impact of handovers on rebuffer in the RAN section (§5.2).

4.3 Broadcaster-to-Viewer Delay

Finally, we study the overall broadcaster-to-viewer (B2V) delay, *i.e.*, the delay between a frame creation at a broadcaster and the playback at a viewer, which significantly impacts the interaction experience between broadcasters and viewers.

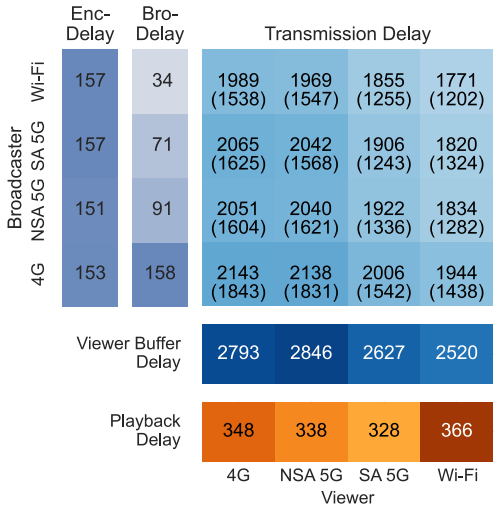
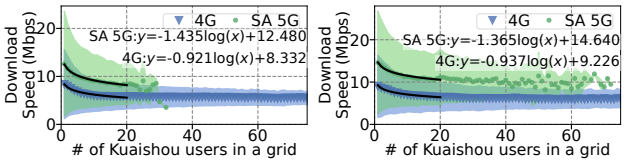


Figure 11: Broadcaster-to-viewer delay decomposition. Delays in brackets correspond to streams without transcoding.

B2V Delay Decomposition. As illustrated in Fig. 10, we first decompose the whole B2V delay into the following delays: 1) *encoding delay*, the delay for the broadcaster to encode a frame; 2) *broadcast delay*, the time a frame being buffered at the broadcaster before being sent to the source server; 3) *transmission delay*, the time for a frame to be delivered to the source servers, transcoded to different bitrates (if needed), delivered to CDN servers, and delivered to a viewer; 4) *viewer buffer delay*, the time a frame being buffered at the viewer before playback; 5) *playback delay*, the delay for a frame to be processed at the viewer's player.

Delay Analysis. In Fig. 11, we plot the above delays. As streams with and without transcoding should have great differences in transmission delays, we distinguish them by bracketing streams without transcoding in the transmission delay block. For the *encoding and playback delay*, they are mainly determined by devices' inherent capacity, and thus similar in all networks. For the *broadcast delay and viewer buffer delay*, Wi-Fi and SA 5G users have shorter delays than NSA 5G and 4G users. This is because, for broadcasters, lower latency and higher bandwidth of Wi-Fi and SA 5G allow the transport protocol to enlarge the congestion window and thus shorten the queuing delay; for viewers, the SA 5G and Wi-Fi users spend less time on rebuffers and thus are able to maintain a smaller buffer length. For the *transmission delay*, using SA 5G at either broadcasters or viewers reduces it, and using SA 5G at both sides leads to 237ms (11%) lower delay than using 4G in streams with transcoding, while 600ms (33%) lower in streams without transcoding.

Given that the transcoding servers do not distinguish traffic from different access networks, it is interesting to find that SA 5G's advantages over 4G for the transmission delay are less in streams with transcoding than in streams without transcoding, in terms of both absolute decrease (237ms *vs.* 600ms) and relative decrease (11% *vs.* 33%). To dig out the reason, we again calculate the distances between users and their serving servers for two kinds of streams separately. We find that in cellular networks, the major differences lie on the broadcaster side: for streams without transcoding, their average user-server distances (320~340km) are much shorter than those of streams with transcoding (580~600km). This is because that streams without transcoding can be uploaded to nearby CDN servers, while streams with transcoding can only be uploaded to certain clusters containing both source servers and transcoding servers, which were only located in less than 5 cities and much more centralized than CDN servers at that time. Therefore, we conclude that *SA 5G can benefit more from distributed servers in terms of delay*. This is consistent with our expectation: the differences between SA 5G and 4G lie on the cellular part, and longer internet travelling distances will weaken SA 5G's superiority. This observation also inspires us to take a deep dive into the relationship between the delay and packets' travelling distances in cellular networks or in the internet by active measurement in §6.



(a) Beijing. (b) Shenzhen.

Figure 12: Download speed versus access density.

5 RADIO ACCESS NETWORK PERFORMANCE

In this section, we focus on the 5G radio access network (RAN). We uncover the impact of access density and handover on crowd-sourced live streaming, as well as energy consumption with the passively collected data.

5.1 SA 5G: More Sensitive to Access Density

We know that SA 5G performs much better than 4G in many metrics, but more people using 5G will deteriorate the download speed. We are still eager to know whether people's gathering directly impacts on its performance in RAN. To find out the answer, we investigate the relationship between download speed and "access density". As the non-5G-oriented API prevents us from conducting cell-wide measurements, we define the access density as the number of Kuaishou users, *which can be regarded as samples of all cellular users*, in a "grid" (*e.g.*, 2.2km×2.2km square). To focus on RAN and avoid the impact of the bottlenecks and fluctuations of the internet backbone and core network, we select only non-peak hours with stable global QoS performance, *i.e.*, 12:00-18:00 each day.

We divide Beijing and Shenzhen into grids to calculate the access density and the average download speed in each ten-minute time window in each grid. In Fig. 12, we plot the average download speed of viewers in all grids versus the access density of each grid. We observe that for both users in 4G and users in SA 5G, the average download speeds drop when there are more users inside that grid until the density achieves 20 per grid. However, when the density is above 20, the average download speeds become relatively stable. To better understand the decreasing trend, we establish a regression in the form of $y = A\log(x) + B$ for density between 1 and 20, with the results labeled in Fig. 12. We find that SA 5G has a larger absolute value of A, meaning it is more sensitive to access density, *e.g.*, download speed in SA 5G decreases by 4.3Mbps as the density increases from 1 to 20 in Beijing, while download speed in 4G

Table 2: # of handovers of Android viewers under China Mobile detected in September 2021.

# of handovers ($\times 10^6$)		handover destination		
		4G	NSA 5G	SA 5G
handover source	SA 5G	3.7	2.3	6.4
	NSA 5G	5.0	0.5	1.4
	4G	23.4	4.5	3.7

only decreases by 2.7Mbps for the same extent increase of density. Considering SA 5G's download speeds are still higher than 4G's, we conclude that *both the advanced technology and fewer users of SA 5G contribute to its better performance.*

5.2 SA 5G: More Tolerant to Horizontal Handovers

In cellular networks, users' mobility causes handovers, which refer to the change of serving base stations. There are two kinds of handovers, *i.e.*, the vertical handover, which means the change of network type, and the horizontal handover, which means the change of base station in one network type. Handovers in 4G and NSA 5G have been demonstrated to impact QoS or QoE extremely [26, 44, 46]. We then investigate handovers in SA 5G, especially the frequency and their impact on rebuffers.

Statistics of Handovers. In our datasets, ~80% of the 4G and NSA 5G logs report valid base station information, while 4% of SA 5G logs report valid and 34% report partly valid (*i.e.*, either LAC or CID is valid) base station information. This would make the number of horizontal handovers under-estimated, but still meaningful for the following analysis.

In Table 2, we calculate the number of different types of handovers detected by Kuaishou for China Mobile users in September 2021. Firstly, we find that in SA 5G and 4G, although being under-estimated, the horizontal handovers are still more than vertical handovers. However, in NSA 5G, vertical handovers are more than horizontal handovers. We infer that this is because the relatively small coverage of NSA 5G base stations and their co-location with 4G base stations make the NSA 5G's coverage not spatially continuous.

Secondly, based on the detected handovers and statistics in §3.3, we find the frequency of horizontal handovers in SA 5G (*i.e.*, $\frac{6.4\times10^6}{8\%\ \#\ \text{cellular logs}}$) is ~3 times of that in 4G (*i.e.*, $\frac{23.4\times10^6}{89\%\ \#\ \text{cellular logs}}$), although much fewer SA 5G logs report valid base station information and handovers in SA 5G are much more severely under-estimated. We assert that users in SA 5G have much more frequent horizontal handovers than in 4G.

Impact of Handover on Rebuffers. We next investigate the impact of handover on rebuffer frequency. For any two consecutive logs, if their network types or base stations are different, there should be a handover. We then calculate the average rebuffer frequency by the former logs as frequency before handover, and the frequency by the latter logs as frequency after handover. The results are presented in Fig. 13, where we also plot the average rebuffer frequencies in different network types as baselines for comparison. We find that the rebuffer frequencies before handovers are already higher than corresponding baselines. We infer that this is because users are at the edge of cellular cells before handover, and thus are

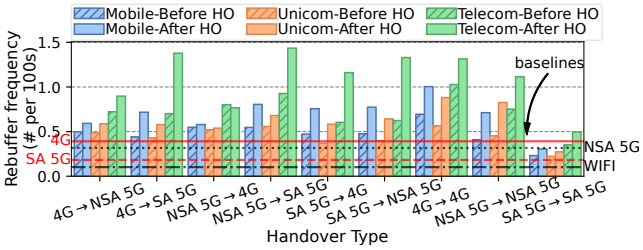


Figure 13: Rebuffer frequencies of Android viewers before and after handover (HO).

likely to suffer from poor signal strength or high interference [27], resulting in higher rebuffer frequencies.

For *vertical handovers*, ranked by the increase in rebuffer frequency by percentage, handovers between NSA 5G and 4G always rank the top 2 in three ISPs, meaning that they cause the least rebuffer frequency increases. Especially, in China Telecom, handovers from NSA 5G to 4G even lead to a 4% drop in the rebuffer frequency. This may be because NSA 5G still uses 4G base stations for the control plane and thus has fewer chances for outages. On the other hand, vertical handovers involved with SA 5G, which increase the rebuffer frequencies by 35%~113%, rank the last 4 in all ISPs. This may be because they have more complex procedures to overcome the "generation gap" on the control plane [7, 35].

For *horizontal handovers*, those in SA 5G increase the rebuffer frequency by 23%~42%, while those in 4G increase by 28%~56% and those in NSA 5G increase by 48%~82%. Therefore, we conclude that SA 5G improves the reliability and has better "handover tolerance" to horizontal handovers.

5.3 SA 5G: Not Always More Power-Consuming

As battery consumption has become an important factor of users' QoE in live streaming [40], we then study the energy consumption for users, using the battery level as an indicator.

Methodology for Fair Comparisons. As different device models differ greatly in the battery capacity, the comparisons of battery levels should only be made within each device model. According to experiential knowledge and features of Li-ion battery [37], the battery level decreases less linearly at the edge. Thus, we filter out those logs with battery level being too high (*e.g.*, >95) or too low (*e.g.*, <10), or battery being charged.

Many factors can influence the energy consumption. From what Kuaishou app could collect, besides ISP and network type, the video bitrate, which impacts the transmission duration and power, the en-/decoding mode (whether using hardware or software), which influences the power of GPU and CPU, and the coding of a stream (*e.g.*, H.264 or H.265), which indicates the complexity of en-/decoding procedure, should all be controlled for fair comparisons. For other factors we could not collect, such as screen brightness and background apps, we assume they follow similar distributions and thus their impacts could be minimized if we collect enough data. Actually, in our experiments, we have selected 22.4 million devices with 29 popular 5G device models, including 27 Android models (19.4 million devices) and 2 iOS models (3 million devices). To fully utilize the data and reduce randomness, we use linear regression

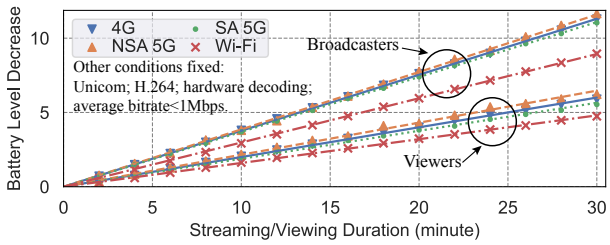


Figure 14: A showcase of battery level decrease for broadcasters and viewers with HUAWEI Mate30 Pro under different networks.

for further comparisons. The following showcase describes the rationale.

A Showcase. We first present a showcase of battery level decrease versus streaming/viewing duration in Fig. 14, with the fixed condition labeled in the figure. This figure shows that the battery level decreases linearly over time in each network for HUAWEI Mate30 Pro. For other devices and settings, the battery level decreases also follow the linear trend well. So that we calculate the *battery level decrease rate* using linear regression, which enables us to make comparisons among different networks or other factors. In fact, in the following results of linear regression, 97.2% of coefficients of determination (R^2) are greater than 0.990, and the minimum value is 0.963. Such high values of R^2 add to the reliability of the conclusions to be drawn. For the following experiments, the settings in the showcase will be inherited.

We want to verify or refute some pieces of conventional wisdom: i) 5G is always more power-consuming than 4G, and NSA 5G is more power-consuming than SA 5G, as NSA 5G requires two connections with both 4G and 5G base stations [11, 21, 44]; ii) Higher radio bands are more power-consuming than lower radio bands in SA 5G [15]; iii) Higher transmission rate mitigates the efficiency gap between SA 5G and 4G [33, 44].

Energy Consumption Across Network Types. For the first piece of conventional wisdom, in Fig. 15(a), we calculate *the battery level decrease rate normalized by that in 4G*, *i.e.*, a "normalized decrease" above 1 suggests the device is more power-consuming in a particular network than in 4G. We first find that devices consume much less (by ~20%) energy in Wi-Fi than in cellular networks. Within cellular networks, we find that Kylin and Dimensity are more SA 5G-friendly, with the differences between SA 5G and 4G ranging from −5% to 6.5%; while Snapdragon and A14 have greater differences ranging up to 18% and 17%, respectively. Meanwhile, among the 17 device models with Kylin and Dimensity 5G SoCs, 7 models are less power-consuming in SA 5G than in 4G. Among all the 29 models, SA 5G are more power-consuming than NSA 5G for 14 models. This phenomenon suggests that *the energy consumption performance is highly related to 5G SoC, and 5G (NSA 5G resp.) is not always more power-consuming than 4G (SA 5G resp.).* The observation that same SoCs perform similarly across different models also indicates that our methodology greatly alleviates the impact of uncontrolled factors.

Energy Consumption Across ISPs. Each of the three ISPs in China has deployed only one major radio band for 5G currently: China Mobile utilizes 2.6GHz while China Unicom and Telecom

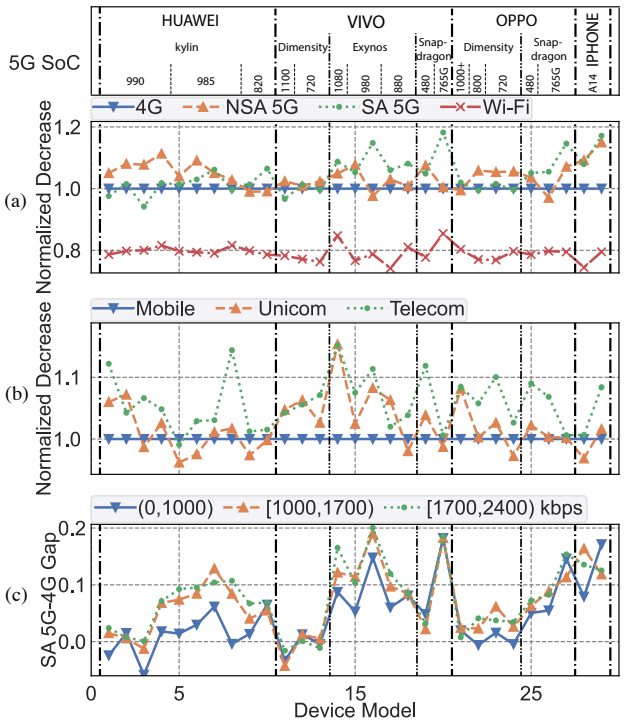


Figure 15: Energy consumption of different device models.

utilize 3.5GHz radio band for 5G. We compare energy consumption among different ISPs to check the second piece of conventional wisdom. As illustrated in Fig. 15(b), each curve shows *the normalized battery level decrease rate of SA 5G against China Mobile.* We observe that for 28 out of 29 device models, they are more power-consuming in China Telecom than in China Mobile (by 0.1%~15.2%), suggesting the radio band plays an important role in power consumption. However, although China Unicom and Telecom co-construct the base stations and share similar radio bands, their energy performances differ greatly, indicating that *the power consumption performance is not merely related to the radio access network.* The second piece of conventional wisdom does not always hold true either.

Energy Consumption Across Video Bitrates. In Fig. 15(c) we study the impact of video bitrate on power consumption to check the third piece of conventional wisdom. The points in this figure are *the SA 5G-4G gap* of different bitrate levels, defined as $(k_{SA} - k_{4G})/k_{4G}$, where k_{SA} and k_{4G} are the battery decrease rate for SA 5G and 4G, respectively. We observe that for 18 out of 29 device models, the lowest bitrate levels correspond to the lowest SA 5G-4G gaps, and as bitrate level increases, the SA 5G-4G gaps increase as well. Therefore, the last piece of conventional wisdom does not hold true as well.

In summary, energy consumption for users using mobile networks in the wild is affected by many factors, and all the above conventional wisdom does not always hold true.

6 CORE NETWORK PERFORMANCE

A major feature of SA 5G deployment is the control and user plane separation (CUPS), which enables the user plane function (UPF)

Table 3: Extracted fields from traceroute result.

Field	Description and example
Domain	Domain name, *e.g.*, "ali.pull.yximgs.com"
ISP	ISP of the user, *e.g.*, "China Mobile"
RTT	RTT between user and server (ms), *e.g.*, "18.96"
u-city	City where the user is, *e.g.*, "Tianjin"
s-city	City of destination server (parsed by IP), *e.g.*, "Beijing"
gw-city	City of gateway (parsed by IP), *e.g.*, "Beijing"
gw-hop	Rank of the hop corresponding to gateway, *e.g.*, "4"

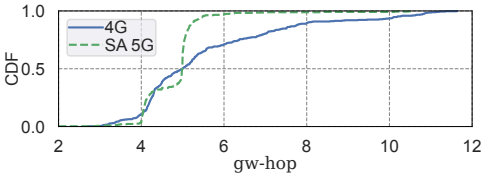


Figure 16: CDFs of the number of traceroute hops to the internet.

to be placed much closer to users [50]. In this section, we want to know whether users are closer to the internet? And if so, whether being closer to the internet helps reduce latency? To answer these questions, we first analyze the traceroute results and extract some useful fields.

For each traceroute task, the domain name, the ISP of the user, and the RTT between the user and the destination server are first extracted. For geolocational analysis, we extract the user's city and the server's city, which is parsed by the resolved IP with datasets from ipip.net. Besides, as have mentioned in §3.2, we just focus on the first internet hop, *i.e.*, the first hop with a public IP, and use it to estimate the gateway (gw for short). We then extract the gateway's city (parsed by IP) and the rank of its hop. We summarize these fields in Table 3.

6.1 SA 5G Users are Closer to Internet

To answer whether users are closer to the internet, we investigate the gateway from two dimensions. The first is to measure the gw-hop, which represents a "network distance" between the user and the internet. However, the 5G users and network deployment can be biased in China, *i.e.*, more users and deployment in the developed cities and fewer in the developing cities or remote districts, which means that 5G users may be geographically closer to internet servers, because internet servers also tend to be deployed more in the developed cities. Inspired by [23], to exclude the impact of biased distribution, we first calculate the average gw-hop in each city for SA 5G and 4G separately, and then filter out those cities that have results obtained only in SA 5G or in 4G. Taking China Mobile users and domain name ks.push.yximgs.com as an example, we plot the CDFs of average gw-hop in Fig. 16.

We find that in SA 5G, more than 97% of cities can reach the internet within 6 hops on average, while the number for 4G is only 71%, indicating that *SA 5G network users are much closer to the internet*. Besides, we find that the SA 5G curve surges steeply at gw-hop= 4 and gw-hop= 5, which means that the estimated gateway regularly appears at the 4th or 5th hop in different paths. This indicates that the estimated gateway refers to a certain component in every path.

Table 4: Average ratio of gateways being in the same city with users.

	u-city=s-city	u-city≠s-city
4G	80.8%	16.5%
SA 5G	87.6%	56.4%

Table 5: Slopes of RTT vs. Distance (negative values are underlined).

network type	Total distance (km)	Slope of RTT vs distance ($\times 10^{-2}$)					
		user-gateway distance			gateway-server distance		
		Mobile	Unicom	Telecom	Mobile	Unicom	Telecom
SA 5G	0-300	0.1480	0.9265	3.1097	1.7084	-0.5507	-1.1154
	300-600	-2.4341	-0.8435	-0.7331	2.8897	1.4122	1.0475
	600-900	-0.3240	-1.3060	-1.2719	0.5634	1.4598	1.4560
	900-1200	-0.8499	-1.4745	-1.7608	0.8987	1.8262	1.6887
	1200-1500	-2.3246	-0.7057	-0.8835	2.3564	0.7005	1.0238
	1500-1800	-1.8692	-1.4516	-1.0007	1.9132	1.4305	1.0332
4G	0-300	3.5890	10.2388	12.7304	-1.2798	-12.0165	-19.2340
	300-600	1.9867	5.1460	5.8469	-1.1730	-5.0031	-4.6184
	600-900	0.9293	2.2684	1.6068	-0.5190	-2.1610	-1.2435
	900-1200	0.8592	1.7962	1.6588	-0.8110	-1.6581	-1.1492
	1200-1500	0.0822	0.1376	0.4374	0.0010	-0.1647	-0.2924
	1500-1800	0.1365	0.5914	0.1430	-0.1049	-0.5774	-0.0776

For the second dimension, we check whether the gateways are in the users' cities. To avoid the similar potential bias from the users' distribution, we first separate the data into two parts by whether the user and the server are in the same city, and then calculate the ratio of gateways being in the same city with the users for each city, before averaging the ratios among all the cities. As shown in Table 4, we observe the ratios for SA 5G are larger than those in 4G, regardless of whether the user and the server are in the same city, meaning that SA 5G has higher chances to serve a user with the local gateway than 4G.

6.2 Shorter Gateway-Server Distance Leads to Lower RTT

We then investigate whether the closer-to-user gateways always help reduce user-to-server RTT, which is related to many QoE metrics. As the RTT is related to the linearized distance of the connecting path [39], in particular, we study the relationship between the RTT and user-gateway (*i.e.*, u-gw) distance and the relationship between RTT and gateway-server (*i.e.*, gw-s) distance, under similar total distances, defined as the sum of u-gw distance and gw-s distance.

In Table 5, we present the slopes of RTT versus u-gw distance and gw-s distance in each distance group, obtained by linear regression. We find that when the total distance is above 300km, RTT in SA 5G is negatively related to u-gw distance but positively related to gw-s distance, while 4G shows nearly the opposite pattern. The result indicates that a packet may travel faster in 5G core network than in the internet. This suggests that in SA 5G if a server is far away from the user, it may even cause a larger RTT to route the user's traffic to the internet by a local gateway, as it shortens the u-gw distance and lengthens the gw-s distance. A better choice would be leaving the traffic in the core network until routing it to the internet at a UPF near the server. On the other hand, if the server is close to the user, SA 5G can benefit more from its local gateway by routing traffic to the internet locally. Such observation

and suggestion for routing are also consistent with the conclusion drawn for the broadcaster-to-viewer delay in §4.3: longer internet distances weaken 5G's advantages.

7 DISCUSSION AND ENHANCEMENT

7.1 Discussion and Guidelines

Discussion on Measurement Results. We first discuss our measurement results and the conclusions.

▷ For the *delay*, we measured the connection establishment delay, broadcaster-to-viewer delay, and traceroute-based RTT. Compared with 4G, both NSA and SA 5G reduce the delays, while SA 5G shows much lower delays than NSA 5G. Meanwhile, with UPFs closer to users, SA 5G can benefit more from the distributed servers.

▷ For the *bandwidth*, we measured the download speed nationwide as well as in a small grid. Although SA 5G has much higher download speeds than 4G, which is still lower than the theoretical capacity, both SA 5G's nationwide and grid-wide download speeds deteriorate as the number of cellular users increases. This leads us to think about another question: With the increasing number of 5G users, does there exist an optimal ratio α (α<1) such that in each grid, the best QoS/QoE could be obtained for every cellular subscriber when α of users connecting to SA 5G and $(1-\alpha)$ of users connecting to 4G? In our dataset, there are not enough users in SA 5G in one grid, so we leave this topic as future work.

▷ For the *reliability*, we measured the overall rebuffer proportion as well as the impact of handovers on rebuffer frequency. Compared with 4G, SA 5G shows much smaller rebuffer proportions and better handover tolerance to horizontal handovers. Meanwhile, NSA 5G may even deteriorate the rebuffer proportion on the basis of 4G. For the handover frequency, we find that the horizontal handovers are much more frequent than vertical handovers for 4G and SA 5G, while users in SA 5G may experience more frequent handovers than in 4G.

▷ For the *energy consumption*, our measurement results refute several pieces of conventional wisdom. The energy consumption performance is highly related to 5G SoC, and SA 5G can be less power-consuming than 4G in some scenarios.

Discussion on Limitations. 1) We do not form strict A/B tests in our measurement, as it would be expensive and difficult in such an online system serving real-world users. Instead, we make full utilization of the massive datasets collected, and build sub-datasets to implement quasi-experiments, which guarantee the reliability of conclusions. 2) Due to either OS's constraints or privacy concerns, there are some other factors in-/outside Kuaishou app we cannot obtain, such as background traffic and apps, and screen brightness. This may have influenced the analysis especially on energy consumption. However, the quasi-experiments and the huge number of users make up for such drawback, which can also be seen from the results obtained: the high coefficients of determination, and the observation that certain series of 5G SoCs perform consistently in different models.

Guidelines for 5G Participants. Based on the above discussions, we provide several guidelines for content providers and other key players in this industry to embrace the 5G era.

▷ For *content providers*, as well as *CDN providers*, from the *delay* performance, we suggest that servers, including source servers, transcoding servers and CDN servers, be placed closer to users, *e.g.*, co-located with UPFs. Also, as more users are able to reach the internet locally in SA 5G, the server selection strategy for users may be revised, and geographical information should be considered a more important factor. That is because, in 4G, cellular traffic from a remote district may have to be routed to the internet in the capital city of the province, causing the geographical information of city granularity useless. While in 5G, local UPFs increase the importance of geographical information.

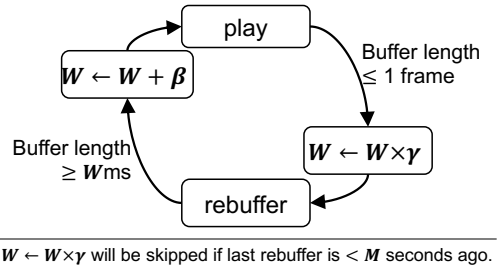


Figure 17: Illustration of rebuffer strategy with adaptive water mark.

From the *bandwidth* performance, as currently the bandwidth is under-utilized, we encourage content providers to provide higher bitrate streams. Besides, the results in the handover part provide content providers the research opportunity for radio access technology (RAT) selection strategy. As vertical handovers to SA 5G always introduce immediate rebuffer probability increase but are expected to reduce the rebuffer in the long run, thus RAT selection decisions should be properly made with user-/app-specific factors, *e.g.*, users' movement pattern, the sessions' expected continuing time, *etc.*

▷ For *ISPs*, to provide lower *delays*, we suggest that the routing policy at UPFs should be carefully designed, *e.g.*, routing traffic with local destination servers to the internet while routing traffic with remote destination servers to another UPF. Also, given that China Unicom and Telecom share base stations and similar radio bands for 5G, their different *reliability* performances indicate there are still optimization opportunities for China Telecom.

▷ For *phone vendors* and *5G SoC manufacturers*, as certain series of 5G SoC have shown better *energy consumption* performance than others, we suggest further optimization could be made to reduce full-scenario SA 5G's energy consumption.

7.2 Potential Improvement: a Trial Study on 5G-aware Rebuffer Strategy

SA 5G has shown better reliability and higher transmission capacity, implying faster outage recovery and smooth playback after recovery. However, we find that the average durations for a single rebuffer in different networks are quite similar, indicating that SA 5G may not perform much better during rebuffers, and the application may underutilize SA 5G. In Kuaishou, the current rebuffer strategy is based on a fixed "water mark" (*e.g.*, 2s): once a rebuffer occurs, the player will keep waiting until the buffer length exceeds the water mark. However, this can be too conservative for SA 5G: once recovering from a rebuffer, SA 5G would again show better

Table 6: Configurations and results of the adaptive water mark strategy in SA 5G.

	baseline	group1	group2
decrease factor γ	-	0.7	0.5
increase step β (ms)	-	500	500
Rebuffer proportion (%)	0.281 (1×)	0.260 (0.93×)	0.269 (0.96×)
B2V delay introduced by rebuffer (ms)	971.3 (1×)	875.6 (0.90×)	928.0 (0.96×)

reliability and higher bandwidth. We deem that an adaptive water mark strategy can better utilize SA 5G's capabilities:

- A water mark W is initialized with the default 100ms;
- W is decreased by a factor γ ($\gamma \in (0, 1)$) once a rebuffer occurs ($W \leftarrow W \times \gamma$), and increased by a step β (in ms) after the playback recovery ($W \leftarrow W + \beta$).
- To avoid frequent rebuffers in poor network conditions, the decrease step is only applied if the last rebuffer occurs more than M (in seconds, *e.g.*, 20s) ago.

This strategy is also illustrated in Fig. 17.

In our A/B tests, besides the baseline group with a fixed water mark, we select two groups of settings on γ and β, as shown in Table 6: *i.e.*, a moderate group and an aggressive group. For each group, we apply the corresponding strategy to ~3 million users for consecutive 18 days in January 2022. As shown in Table 6, compared with the baseline, our strategy can reduce the rebuffer proportion by 4%~7% in SA 5G, as well as the broadcaster-to-viewer delay introduced by rebuffers by 4%~10%. This trial experiment demonstrates that the optimization in application level is useful for better utilization of 5G superiority, and such optimization direction is practical for content providers.

8 RELATED WORK

8.1 5G Mobile Network

Several optimizations and measurement studies have been conducted on 5G networks. Parvez *et al.* [34] categorized the new technologies into three parts: RAN optimization, core network optimization and caching optimization. Based on field tests in the campus, Xu *et al.* [44] found that although 5G networks are expected to provide improved throughput and reduced delay, their coverage, indoor-outdoor gap, and handover performance under the first-stage deployment in 2019, did not have too much advantage against 4G. Narayanan *et al.* [31] conducted an active measurement on NSA 5G in the US by field tests, and found that mmWave 5G is sensitive to the change of environment (*e.g.*, obstruction in line of sight) and the superiority of 5G on web browsing performance is highly related to the page size. Narayanan *et al.* [33] later measured the low, middle and high radio band 5G performance, including both SA (only at low band) and NSA; in particular, they showed that existing adaptive bitrate video streaming strategies underperform in mmWave 5G networks as a result of inaccurate bandwidth prediction.

Narayanan *et al.* [32] aimed to predict the throughput of mmWave in 5G networks, and proposed a machine learning framework using factors including physical-layer information and contextual factors as inputs to predict 5G throughput. Li *et al.* [27] showed that 5G network reliability is not always positively related to received signal strength, and they proposed a signal strength-aware 4G and 5G RAT selection strategy. Xie *et al.* [43] showed that using a physical-layer informed transport protocol can help detect network bottlenecks, guiding TCP congestion control.

8.2 Crowdsource Live Stream

Crowdsource live streaming has been increasingly popular because mobile networks are available to more and more people. Zhang *et al.* [47] studied user behaviors including the viewing patterns and popularity of streams. Xu *et al.* [45] conducted a detailed measurement study of a wide cross-section of popular streaming video-on-demand (VOD) services to develop a holistic understanding of these services' design and performance. Zhu *et al.* [49] established a framework to measure the upstreaming ingest path. For improving quality of experience in video streaming, Ray *et al.* [36] provided a live-streaming upload solution that improves the overall quality of experience for diverse time-shifted viewers by using selective quality-enhancing retransmissions.

9 CONCLUDING REMARKS

With the growing popularity of crowdsourced live streaming and the increasing SA 5G's deployment and the number of its subscribers, we conduct a large-scale measurement study on the QoS/QoE metrics of commercial 5G from a content provider's perspective. Based on one of the largest crowdsourced live streaming platforms, we develop a slice-based data collection mechanism without affecting users' quality of experience. With both passively collected data from Kuaishou app and active measurements by traceroute, we comprehensively investigate the end-to-end performance, the radio access network and core network performance, focusing on four major QoS/QoE concerns: delay, bandwidth, reliability and energy consumption. We also provide insights and guidelines for content providers as well as other 5G participants. A simple but effective rebuffer strategy is proposed and helps reduce rebuffer proportion by 7% in SA 5G in a real-world deployment, providing content providers with the feasibility of further application-level optimization.

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